

SSBMF Status Update: Model Refinement, Validation, and Biological Interpretation (August 3, 2026)

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Summary

- **Selected model:** YFB ($\eta = (YF)\hat{\beta}$), with DeSurv-aligned gene selection, $K = 7$ latent programs, and no cohort indicator.
- **Selected factors:** the model uses **2 of its 7 programs** for survival prediction ($K_{\text{eff}} = 2$) — matching DeSurv's own ~ 1 prognostic factor, and robust across independent checks, not a tuning artifact.
- **Performance:** **mean external C-index = 0.627 across 5 held-out PDAC cohorts**, significantly more concordant than a two-step baseline when pooled across cohorts ($+0.042$, 95% CI 0.013–0.071).
- **Other key findings:** the joint model's advantage over a two-step (factorize-then-Cox) baseline widens as survival signal strength increases; the two survival-active programs are biologically distinct and correspond to the established PDAC adverse (basal-like) / protective (classical) subtype axis, confirmed by five independent methods.

Recent Progress

- **Objective put on a principled footing.** The genomics/survival scale imbalance the lab flagged was addressed; the redundant λ mixing parameter was retired (α already plays its role). Honest finding: for YFB this gave **no performance change** — its genomics half and survival half never share a coordinate needing rebalancing. A separate train/test preprocessing bug (external cohorts were rank-transformed while training was per-platform z-standardized — the reverse) was fixed; **this fix alone** explains the headline moving from 0.636 to **0.627** (a correction, not a regression).
- **Why 7 factors when DeSurv used 3?** Cross-validation genuinely selects $K = 7$, but only **2 factors are survival-active** ($K_{\text{eff}} = 2$, defined as $|\hat{\beta}_k|$ exceeding a fixed threshold of 0.001) — matching DeSurv's own ~ 1 prognostic factor. Two independent optimization strategies (warm-start from a higher- K fit, and a joint (K, α) Bayesian-optimization search) both land on $K_{\text{eff}} = 2$, and a deflation-initialization test analytically ruled out initialization as the cause of an earlier ambiguity; $K \in \{4, 5, 7, 8\}$ are statistically indistinguishable (spread 0.0015). $K = 7$ is kept only for one-step reproducibility; $K = 4/K = 5$ are validated equivalents.
- **Study-specific baseline hazard.** Implemented the requested `+ strata(study)` as an optional stratified Cox partial likelihood. Performance-neutral ($0.6267 \rightarrow 0.6263$), so available but off by default.
- **Biological interpretation of the two active programs.** Complete, and confirmed by five independent methods.

Validation across simulated and real data

Simulation — joint model vs. a two-step (factorize-then-Cox) baseline (re-confirmed under the corrected model this cycle):

- *Turn survival signal up from zero:* at zero signal, joint $\approx$ two-step (both near chance); as signal grows, the joint model pulls ahead and the gap widens, in 3 of 4 data-generating designs. One artificial design (exactly-sparse, symmetric factor loadings) reverses this — traced to a specific, understood factor-collapse failure mode, not a survival-coupling defect.
- *Shared vs. study-specific factors:*

Table 1: Held-out C-index by cross-cohort factor structure (5 seeds).

Scenario	Joint (YFB)	Two-step (EBMF+Cox)
All factors shared	0.871	0.856
Hybrid (some shared, some specific)	0.815	0.717
No shared prognostic factors	0.549	0.550

When no shared prognostic signal exists, joint and two-step are equivalent; in the realistic hybrid case (only some factors prognostic) the joint model leads — as much through **stability** as accuracy: the unsupervised two-step baseline collapses on 2 of 5 seeds (dragging its mean to 0.717), while the joint model is stable across all seeds (mean 0.815) and classifies each factor as shared vs. study-specific with perfect accuracy.

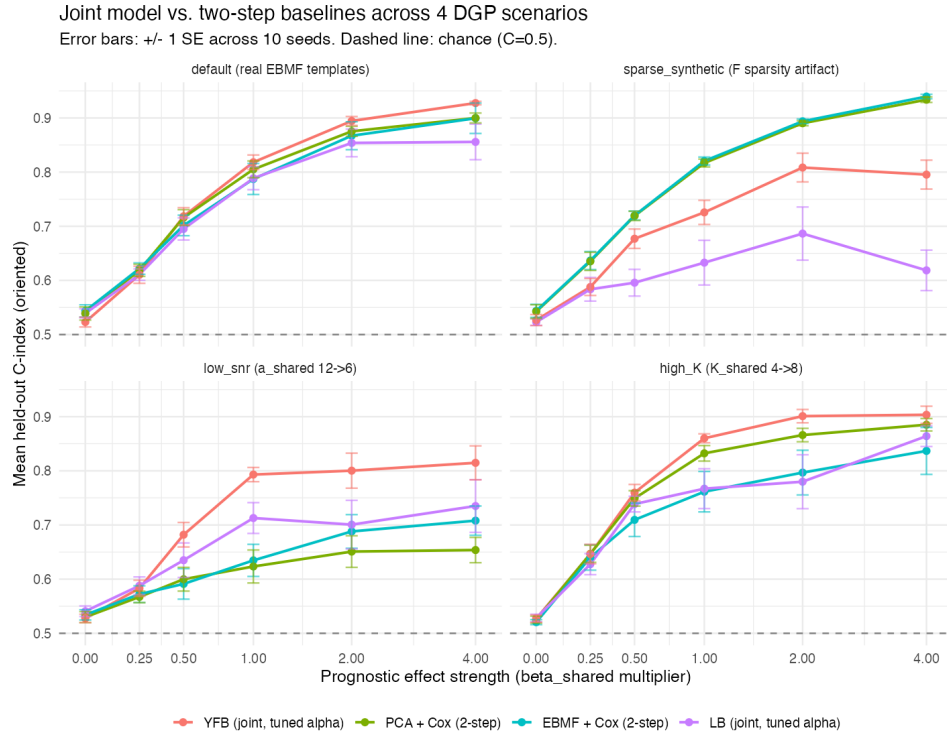


Figure 1: Each of the 4 panels is a different simulation design (default = realistic factor-loading templates; sparse_synthetic = an artificial exact-sparse stress test; low_snr = weakened shared genomic signal; high_K = more shared latent factors) – the panels are not different models. Within every panel, the x-axis is prognostic effect strength and the 4 colored lines are the models being compared: YFB (recommended, joint) and LB (the other joint variant) against two two-step baselines, PCA+Cox and EBMF+Cox. YFB pulls ahead of both two-step baselines as signal strength increases in 3 of 4 panels, with the gap widening through the highest signal level tested; the one reversal (sparse_synthetic) is the known factor-collapse artifact noted above, not a general failure of the joint approach.

Real data — 5 held-out PDAC cohorts (RNA-seq, microarray, proteomics):

Table 2: External validation with 95% bootstrap CI (mean = 0.627).

Held-out cohort	External C-index	95% CI
PACA_AU_seq	0.657	(0.544, 0.772)
PACA_AU_array	0.648	(0.553, 0.747)
Puleo_array	0.645	(0.599, 0.688)
Dijk	0.634	(0.555, 0.708)
Moffitt_GEO_array	0.549	(0.467, 0.625)

Four of five cohorts clear chance with their CI lower bound above 0.5; **Moffitt microarray does not** (CI 0.467–0.625), confirming quantitatively what was previously only a qualitative flag. **Against a two-step (unsupervised EBMF then Cox) baseline on the same patients, a paired bootstrap finds the recommended model significantly more concordant when pooled across all 5 cohorts: +0.042 (95% CI 0.013–0.071).** Individually only the largest cohort (Puleo, $n = 288$) reaches significance alone; the other four all favor the recommended model but are underpowered on their own. Per-platform z-standardization is essential — most non-per-platform preprocessing collapses the survival coefficients to zero.

Bootstrap method: both the per-cohort CIs above and the paired comparison use patient-level bootstrap resampling within each cohort (2000 replicates, percentile CI). The paired comparison resamples patients once per replicate and scores both models on that identical draw, preserving the correlation between their errors on shared patients — a more powerful (and correct) test than comparing two independently-computed CIs. The 5 per-cohort differences are then combined via a fixed-effect, inverse-variance-weighted pooled estimate.

Biology of the two active programs

An **adverse** program (basal-like / squamous / MET-EGFR-driven) and a **protective** program (classical / differentiated-epithelial) — the established PDAC subtype-survival axis. Confirmed by five independent methods: gene-set enrichment, PurlST (Purity Independent Subtyping of Tumors; Rashid et al. 2020) subtype concordance ($\rho = \pm 0.58$), external-cohort hazard ratios (adverse HR > 1 in 5/5 cohorts, protective HR < 1 in 5/5), and direct gene-list overlap with DeSurv’s published programs.

Methods for the five checks above:

- **Gene-set enrichment** — fgsea, ranked by each program’s posterior mean gene weight (the model’s own $E[F]$ column, no re-ranking), tested against MSigDB Hallmark/Reactome/KEGG/GO:BP plus a custom PDAC gene-set collection (Moffitt basal/classical, Bailey 2016, DeSurv D1–D3); ORA on the top 50/100/150-weighted genes serves as a confirmatory cross-check. The figure below pools the fgsea-defined **leading-edge genes** (the genes actually driving each set’s enrichment score, up to 20 per set) across every significant set ($\text{padj} < 0.10$) for Programs 3 and 7; heatmap rows are hierarchically clustered (Euclidean distance, complete linkage) on their weight profile across all 7 programs, columns are fixed in program order, and color is the raw posterior mean weight.
- **PurlST concordance** — Spearman correlation between each patient’s posterior mean program loading and their PurlST basal-likelihood probability (TCGA-PAAD, the only cohort with PurlST calls available).
- **External-cohort hazard ratios** — a univariate Cox model, per cohort, of a leading-edge gene-signature score (mean per-gene z-score across that program’s leading-edge genes) against overall survival.
- **DeSurv gene overlap** — hypergeometric test and Jaccard index between each program’s top-270-weighted genes and DeSurv’s published per-factor gene lists.

Comparison against DeSurv

- **Factor count:** SBMF’s CV-selected $K = 7$ collapses to $K_{\text{eff}} = 2$ survival-active programs — closely matching DeSurv’s own ~1 prognostic factor, via two independent confirmation strategies (see Recent Progress above).
- **Gene selection overlap:** Program 7 (Adverse) overlaps DeSurv’s basal-like-tumor factor at Jaccard 0.18 (hypergeometric $p \approx 3 \times 10^{-16}$); Program 3 (Protective) overlaps DeSurv’s classical-tumor factor at Jaccard 0.13 ($p \approx 1 \times 10^{-7}$). Despite different model classes (empirical-Bayes factorization + Cox vs. DeSurv’s own framework), both methods converge on the same underlying tumor biology — that convergence, not a performance ranking, is the meaningful point of comparison here.

Key next steps

1. Characterize factor-stability across random seeds/initializations for the recommended YFB fit (not yet done).
2. Run a matched, same-protocol head-to-head against DeSurv (same data, same scoring), building on the gene-overlap comparison above.
3. Shift focus to manuscript writing (introduction and background/literature research) — the companion manuscript repository is already scaffolded with an intro draft and populated reference library; the next step is to continue and expand that draft.

